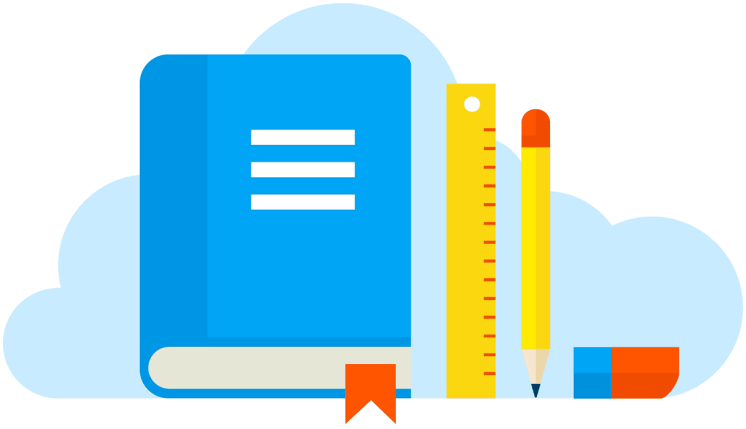


📘 NOTE 7 – INTRODUCTION TO COMPUTER NETWORKS

Date: 19/04/2026

INTRODUCTION TO COMPUTER NETWORKS

Computer Networks Notes

Subject: Computer Networks
Topic: Introduction to Computer Networks
Level: BCA Semester 5

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1. What is a Computer Network?

A **computer network** is a collection of two or more computers and devices connected together to communicate and share resources.

A network allows users to share:

- Data
- Files
- Printers
- Internet connection
- Applications
- Other network resources

Simple Example

In a college computer laboratory, multiple computers can be connected to the same network. Students can access shared resources and communicate with other computers.

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2. Components of a Network

A computer network mainly consists of:

1. Sender

The device that sends data.

2. Receiver

The device that receives data.

3. Transmission Medium

The path through which data travels.

Examples:

- Ethernet cable
- Fiber optic cable
- Wi-Fi

4. Network Devices

Devices that help connect and manage computers.

Examples:

- Router
- Switch
- Hub
- Modem
- Access Point

5. Protocols

Rules used for communication between devices.

Examples:

- HTTP
- TCP
- IP
- FTP
- DNS

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3. Types of Networks

Networks can be classified according to their geographical coverage.

PAN — Personal Area Network

A PAN covers a very small area around an individual.

Example: Bluetooth connection between a smartphone and wireless headphones.

LAN — Local Area Network

A LAN covers a small geographical area such as:

- Home
- Office
- School
- Computer laboratory

LAN usually provides high-speed communication.

MAN — Metropolitan Area Network

A MAN covers a larger area such as a city or metropolitan region.

WAN — Wide Area Network

A WAN covers a very large geographical area.

Example: The Internet.

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4. Network Topology

Network topology refers to the arrangement of devices in a network.

Common types include:

Bus Topology

All devices are connected to a single main cable.

PC —— PC —— PC —— PC

Main Cable

Star Topology

All devices are connected to a central device such as a switch.

```
    PC
     |
PC ── Switch ── PC
     |
    PC
```

Star topology is commonly used in modern LANs.

Ring Topology

Each device is connected to two neighboring devices, forming a ring.

Mesh Topology

Devices have multiple connections with other devices.

It provides high reliability but requires more cables and resources.

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5. Network Devices

Hub

A hub connects multiple devices and broadcasts received data to connected devices.

Switch

A switch connects devices in a LAN and forwards data to the appropriate destination.

Router

A router connects different networks and determines suitable paths for data.

Modem

A modem converts signals to enable communication over certain transmission systems and commonly provides connectivity to an Internet service.

Access Point

An access point allows wireless devices to connect to a network.

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6. Network Protocol

A **protocol** is a set of rules that controls communication between devices on a network.

Important protocols include:

HTTP

HyperText Transfer Protocol is used for communication between web browsers and web servers.

HTTPS

HTTPS is the secure version of HTTP and uses encryption to protect communication.

FTP

File Transfer Protocol is used to transfer files between systems.

TCP

Transmission Control Protocol provides reliable and ordered delivery of data.

IP

Internet Protocol is responsible for addressing and routing packets.

DNS

Domain Name System translates domain names into IP addresses.

For example:

example.com

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IP Address

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7. IP Address

An **IP address** is a numerical address assigned to a device on a network.

Example of an IPv4 address:

192.168.1.10

An IP address helps identify and communicate with a device on a network.

IPv4

IPv4 uses **32-bit addresses**.

Example:

192.168.1.10

IPv6

IPv6 uses **128-bit addresses** and was developed to provide a much larger address space.

Example:

2001:db8::1

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8. OSI Model

The **OSI (Open Systems Interconnection) model** divides network communication into seven layers.

The seven layers are:

7. Application

6. Presentation

5. Session

4. Transport

3. Network

2. Data Link

1. Physical

Easy way to remember:

A P S T N D P

All People Seem To Need Data Processing

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9. Transport Layer

The Transport Layer provides communication between applications running on different devices.

Two important protocols are:

TCP

- Reliable
- Connection-oriented

- Ensures ordered delivery
- Performs error and flow control

UDP

- Faster and lightweight
- Connectionless
- Does not guarantee delivery
- Useful where speed is important

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10. Network Security

Network security protects network resources and data from unauthorized access, attacks, and misuse.

Common security techniques include:

- Firewall
- Encryption
- Authentication
- Antivirus software
- Access control
- Secure protocols

Firewall

A firewall monitors and controls network traffic according to defined security rules.

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11. Advantages of Computer Networks

Computer networks provide many benefits:

- Resource sharing
- File sharing

- Communication
- Internet sharing
- Centralized management
- Faster information exchange
- Reduced operational cost

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12. Computer Networks in StudySphere

Networking is important for a web-based application such as StudySphere.

The basic communication flow is:

Student Browser

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Internet / Network

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StudySphere Server

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Database

For example, when a student opens the StudySphere Notes page, the browser sends a request to the server. The server processes the request and returns the required information.